

Predicting the next credit-risk class

A longitudinal multiclass classification project built in R

R • Random Forest • XGBoost • LightGBM • Elastic Net

12,500

customers

100,000

monthly records

The project in one view

Business question

Can we predict each customer's next monthly credit class — Bad, Standard or Good — from current behaviour and prior score history?

Result

A cost-aware, customer-level validation pipeline selected an ensemble for one-step prediction and a Random Forest-assisted autoregressive setup for multi-month forecasting.

329

model candidates

69.2%

selected setup accuracy

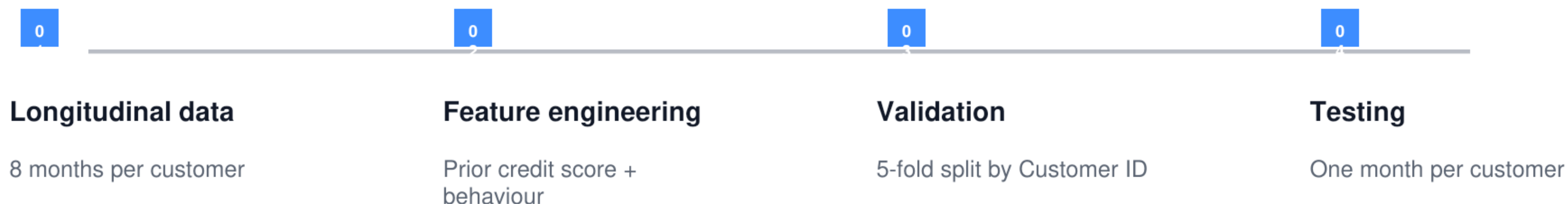
75.1%

first-month accuracy

68.2%

reported 8-month cumulative

Customer-level splitting prevents identity leakage



Why it matters

All rows from the same customer stay within one fold, so validation measures generalisation to unseen customers rather than memorisation of repeated identities.

Breadth first, then disciplined model selection

01

Elastic Net

interpretable baseline

02

Random Forest

non-linear interactions

03

XGBoost

boosted decision trees

04

LightGBM

efficient gradient boosting

4

base models

2

blending models

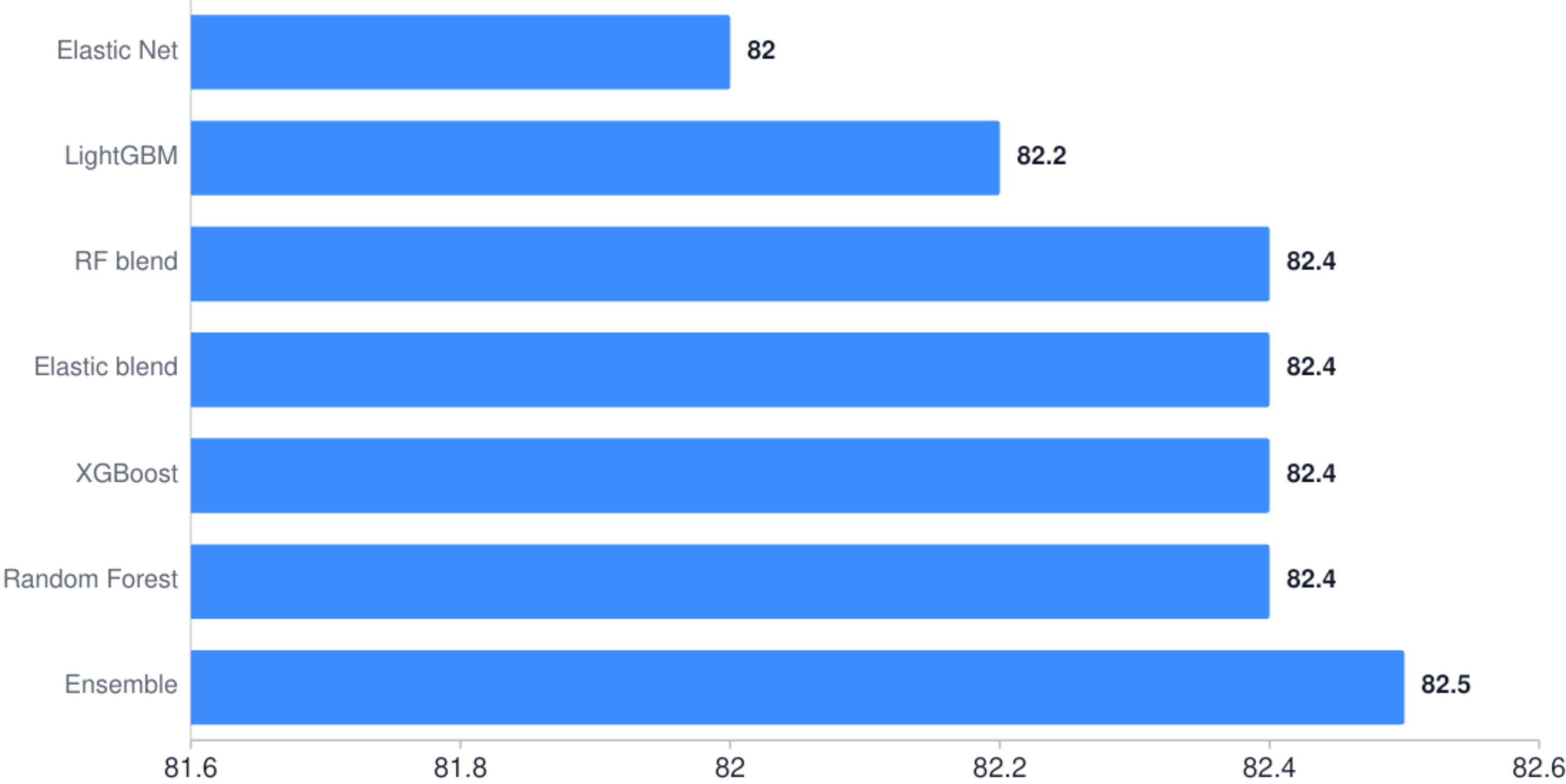
323

weighted ensembles

329

total candidates

Ensemble leads the benchmark



82.5%

best reported accuracy in the one-step benchmark

Selection logic

Accuracy was considered together with an error-direction ratio designed to expose costly optimistic classifications.

Not all credit errors carry the same risk

A Standard accuracy

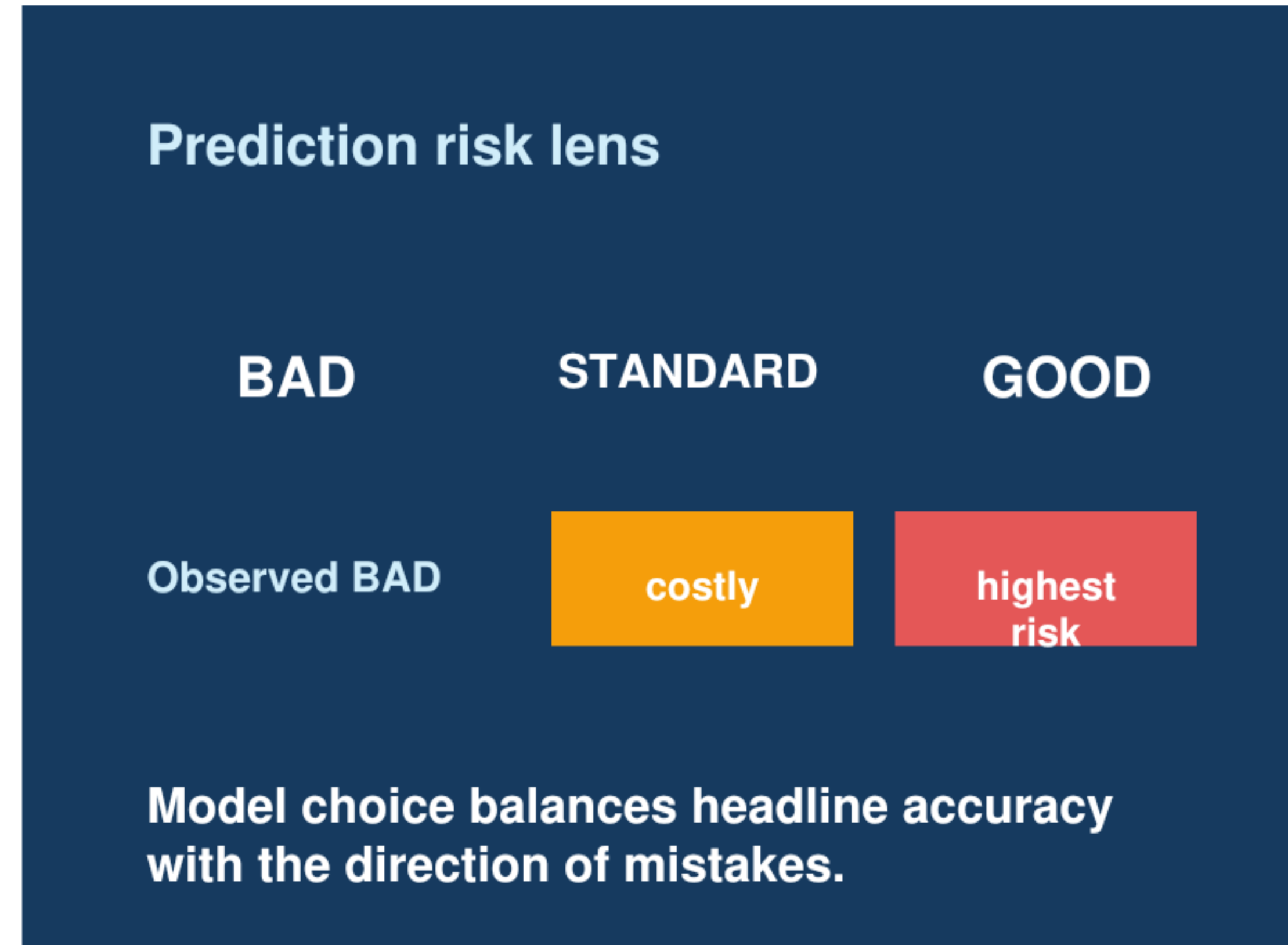
Measures total correct classifications across Bad / Standard / Good.

B Cost-weighted accuracy

The implementation multiplies Bad → Good errors by 5 to reflect higher downside.

C Upper-side error ratio

Tracks errors that predict a better class than the observed outcome.



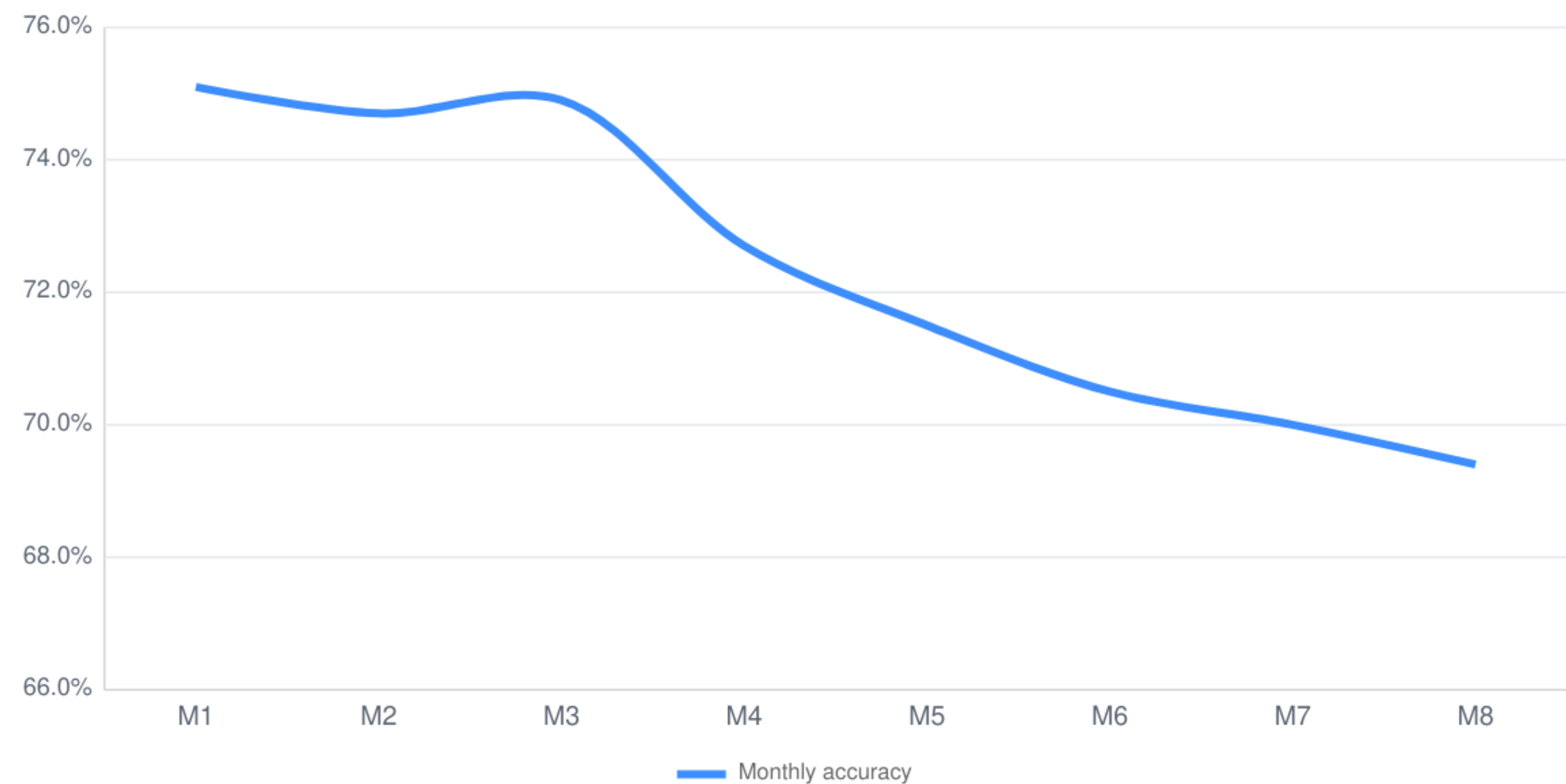
Multi-month forecasts feed predictions forward



Technical achievement

The pipeline explicitly tests how missing history, error propagation and feature reliance affect sequential predictions.

Accuracy declines as forecast distance grows



75.1%

first-month accuracy

68.2%

reported 8-month cumulative accuracy

Interpretation

Longer horizons compound earlier classification errors; production use should monitor performance by horizon.

I can turn an ambiguous risk problem into a validated, business-aware machine-learning workflow.

Analytical judgement

Customer-level validation, multiple metrics and explicit horizon analysis.

Technical execution

R pipeline across Elastic Net, Random Forest, XGBoost, LightGBM and ensembles.

Business communication

Risk-aware evaluation translates model errors into decision-relevant trade-offs.

Next step: productionise with time-based holdout validation, class-level recall, calibration and fairness monitoring.